

Part E - Information Retrieval

Assignment - 1

Shaurya Sharma - 2410110614

Suvigya Khare - 2410110482

1) Free text queries:

Query	Docs (By Rank)	Cosine Scores
cotton shirt	D001,D041,D061,D081,D022,D082,D042,D002,D062,D021	0.2281,0.2281,0.2281,0.2264,0.2262,0.2262,0.2252,0.2235,0.2235,0.2229
denim jeans	D023,D083,D043,D003,D063,D053,D013,D073,D033,D093	0.2577,0.2577,0.2565,0.2532,0.2532,0.2184,0.2182,0.2182,0.2155,0.2155
winter wear	D058,D018,D078,D038,D098,D010,D070,D028,D088,D030	0.1882,0.1872,0.1872,0.1857,0.1857,0.1265,0.1265,0.1264,0.1264,0.1259
festive kurta	D004,D064,D024,D084,D034,D094,D054,D014,D074,D044	0.1862,0.1862,0.185,0.185,0.1847,0.1847,0.1836,0.1826,0.1826,0.181

breathable kurta	D004,D064,D024,D084,D034,D094,D054,D014,D074,D044	0.2249,0.2249,0.2234,0.2234,0.2231,0.2231,0.2216,0.2205,0.2205,0.2185
slim fit jeans	D053,D013,D073,D033,D093,D023,D083,D043,D003,D063	0.275,0.2748,0.2748,0.2714,0.2714,0.1781,0.1781,0.1773,0.1751,0.1751
zip closure jacket	D058,D018,D078,D028,D088,D038,D048,D098,D008,D068	0.1086,0.1081,0.1081,0.1078,0.1078,0.1072,0.1072,0.1064,0.1064,0.1064
high waist denim	D029,D049,D089,D009,D069,D019,D059,D079,D039,D099	0.224,0.224,0.224,0.2187,0.2187,0.1518,0.1518,0.1518,0.1483,0.1483
cotton stretch shirt	D029,D049,D089,D009,D069,D053,D001,D041,D061,D081	0.1663,0.1663,0.1663,0.1624,0.1624,0.1482,0.1459,0.1459,0.1459,0.1449
cotton polyester	D028,D088,D048,D008,D068,D001,D041,D061,D081,D021	0.1243,0.1243,0.1237,0.1227,0.1227,0.0322,0.0322,0.0319,0.0314,0.0314
xyzpolester	-	-

A few screenshots:

Search the clothing corpus using Vector Space Model or positional search.

Select search mode:

☒ Free Text (VSM) ☐ Phrase Search ☐ Proximity Search

Free-Text Ranked Retrieval

Enter your clothing query:

cotton shirt

Uses Inc.Itc cosine similarity and returns the top 10 results.

Top 10 Results

Rank	DocID	Title	Category	Cosine Score
1	D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	0.2281
2	D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	0.2281
3	D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	0.2281
4	D081	Men's Cotton Crew Neck T-Shirt - Mustard	T-Shirt	0.2264
5	D022	Men's Checked Cotton Shirt - White	Shirt	0.2262
6	D082	Men's Checked Cotton Shirt - Blue	Shirt	0.2262
7	D042	Men's Checked Cotton Shirt - Maroon	Shirt	0.2252
8	D002	Men's Checked Cotton Shirt - Black	Shirt	0.2235
9	D062	Men's Checked Cotton Shirt - Olive	Shirt	0.2235
10	D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	0.2229

Search the clothing corpus using Vector Space Model or positional search.

Select search mode:

☒ Free Text (VSM) ☐ Phrase Search ☐ Proximity Search

Free-Text Ranked Retrieval

Enter your clothing query:

festive kurta

Uses Inc.Itc cosine similarity and returns the top 10 results.

Top 10 Results

Rank	DocID	Title	Category	Cosine Score
1	D004	Men's Regular Fit Kurta - Pink	Kurta	0.1862
2	D064	Men's Regular Fit Kurta - White	Kurta	0.1862
3	D024	Men's Regular Fit Kurta - Teal	Kurta	0.185
4	D084	Men's Regular Fit Kurta - Beige	Kurta	0.185
5	D034	Women's Printed Straight Kurta - Maroon	Kurta	0.1847
6	D094	Women's Printed Straight Kurta - Teal	Kurta	0.1847
7	D054	Women's Printed Straight Kurta - Mustard	Kurta	0.1836
8	D014	Women's Printed Straight Kurta - Beige	Kurta	0.1826
9	D074	Women's Printed Straight Kurta - Pink	Kurta	0.1826
10	D044	Men's Regular Fit Kurta - Navy Blue	Kurta	0.181

Free-Text Ranked Retrieval

Enter your clothing query:

xyzpolyester

Uses Inc.Itc cosine similarity and returns the top 10 results.

No results found. The query may contain terms that are not present in the corpus or have zero IDF.

2) Exact phrase queries:

Phrase Query	Docs (by rank)	Ending Positions
cotton shirt	D002,D022,D042,D062,D082	[3, 8], [3, 8], [3, 8], [3, 8], [3, 8]
Stretch denim	D003,D013,D033,D043,D063,D073,D093	[16],[15],[16],[15],[16],[15],[16]
Festive wear	D005,D015,D025,D035,D045,D055,D065,D075,D085,D095	[26],[24],[26],[24],[26],[24],[26],[24],[26],[24]
Winter wear	D008,D010,D018,D020,D028,D030,D038,D040,D048,D050	[26],[23],[25],[27],[26],[23],[25],[25],[26],[25]
Regular fit	D003,D004,D020,D023,D024,D040,D043,D044,D060,D063	[2, 8], [2, 7], [2, 8], [2, 8], [2, 7], [2, 7], [2, 8], [2, 8], [2, 7], [2, 8]
cottonxyz	-	-

A few screenshots:

Clothing Information Retrieval System

Search the clothing corpus using Vector Space Model or positional search.

Select search mode:

☐ Free Text (VSM) ☒ Phrase Search ☐ Proximity Search

Phrase Search

Enter a phrase:

cotton shirt

Finds documents where the query terms occur consecutively using the positional index.

Matching Documents

Rank	DocID	Title	Category	Matching Ending Positions
1	D002	Men's Checked Cotton Shirt - Black	Shirt	[3, 8]
2	D022	Men's Checked Cotton Shirt - White	Shirt	[3, 8]
3	D042	Men's Checked Cotton Shirt - Maroon	Shirt	[3, 8]
4	D062	Men's Checked Cotton Shirt - Olive	Shirt	[3, 8]
5	D082	Men's Checked Cotton Shirt - Blue	Shirt	[3, 8]

Positional index used successfully. The positions shown are the ending positions of matching phrase occurrences.

Deploy

Phrase Search

Enter a phrase:

festive wear

Finds documents where the query terms occur consecutively using the positional index.

Matching Documents

Rank	DocID	Title	Category	Matching Ending Positions
1	D005	Women's Printed Daily Wear Saree - Blue	Saree	[26]
2	D015	Women's Cotton Handloom Saree - Red	Saree	[24]
3	D025	Women's Printed Daily Wear Saree - Maroon	Saree	[26]
4	D035	Women's Cotton Handloom Saree - Yellow	Saree	[24]
5	D045	Women's Printed Daily Wear Saree - Purple	Saree	[26]
6	D055	Women's Cotton Handloom Saree - Pink	Saree	[24]
7	D065	Women's Printed Daily Wear Saree - Green	Saree	[26]
8	D075	Women's Cotton Handloom Saree - Blue	Saree	[24]
9	D085	Women's Printed Daily Wear Saree - Red	Saree	[26]
10	D095	Women's Cotton Handloom Saree - Maroon	Saree	[24]

Positional index used successfully. The positions shown are the ending positions of matching phrase occurrences.

Phrase Search

Enter a phrase:

cottonxyz

Finds documents where the query terms occur consecutively using the

No exact phrase matches found.

3) Proximity Queries:

Proximity Query	Docs (by rank)	Position Pairs
cotton WITHIN/3 shirt	D001,D002,D021,D022,D041,D042,D061,D062,D081,D082	[(16, 19)], [(2, 3), (7, 8)], [(18, 21)], [(2, 3), (7, 8)], [(16, 19)], [(2, 3), (7, 8)], [(16, 19)], [(2, 3), (7, 8)], [(16, 19)], [(2, 3), (7, 8)]
stretch WITHIN/4 denim	D003,D013,D033,D043,D063,D073,D093	[(15, 16)], [(14, 15)], [(15, 16)], [(14, 15)], [(15, 16)], [(14, 15)], [(15, 16)]
winter WITHIN/3 wear	D008,D010,D018,D020,D028,D030,D038,D040,D048,D050	[(25, 26)], [(22, 23)], [(24, 25)], [(26, 27)], [(25, 26)], [(22, 23)], [(24, 25)], [(24, 25)], [(25, 26)], [(24, 25)]
festive WITHIN/4 wear	D005,D015,D025,D035,D045,D055,D065,D075,D085,D095	[(25, 26)], [(23, 24)], [(25, 26)], [(23, 24)], [(25, 26)], [(23, 24)], [(25, 26)], [(23, 24)], [(25, 26)], [(23, 24)]

A few screenshots:

Proximity Search

Enter a proximity query:

cotton WITHIN/3 shirt

Format: term1 WITHIN/k term2

Matching Documents

Rank	DocID	Title	Category	Matching Position Pairs
1	D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[(16, 19)]
2	D002	Men's Checked Cotton Shirt - Black	Shirt	[(2, 3), (7, 8)]
3	D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[(18, 21)]
4	D022	Men's Checked Cotton Shirt - White	Shirt	[(2, 3), (7, 8)]
5	D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[(16, 19)]
6	D042	Men's Checked Cotton Shirt - Maroon	Shirt	[(2, 3), (7, 8)]
7	D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	[(16, 19)]
8	D062	Men's Checked Cotton Shirt - Olive	Shirt	[(2, 3), (7, 8)]
9	D081	Men's Cotton Crew Neck T-Shirt - Mustard	T-Shirt	[(16, 19)]
10	D082	Men's Checked Cotton Shirt - Blue	Shirt	[(2, 3), (7, 8)]

Positional index used successfully. Each position pair represents the positions of the two query terms.

Proximity Search

Enter a proximity query:

winter WITHIN/3 wear

Format: term1 WITHIN/k term2

Matching Documents

Rank	DocID	Title	Category	Matching Position Pairs
1	D008	Women's Casual Puffer Jacket - Olive	Jacket	[(25, 26)]
2	D010	Women's Fleece Sweatshirt - Black	Sweatshirt	[(22, 23)]
3	D018	Women's Quilted Winter Jacket - Beige	Jacket	[(24, 25)]
4	D020	Men's Regular Fit Sweatshirt - Navy Blue	Sweatshirt	[(26, 27)]
5	D028	Women's Casual Puffer Jacket - Black	Jacket	[(25, 26)]
6	D030	Women's Fleece Sweatshirt - Maroon	Sweatshirt	[(22, 23)]
7	D038	Women's Quilted Winter Jacket - Olive	Jacket	[(24, 25)]
8	D040	Men's Regular Fit Sweatshirt - Black	Sweatshirt	[(24, 25)]
9	D048	Women's Casual Puffer Jacket - Beige	Jacket	[(25, 26)]
10	D050	Women's Fleece Sweatshirt - Navy Blue	Sweatshirt	[(24, 25)]

Positional index used successfully. Each position pair represents the positions of the two query terms.

Effect of Positional Information:

Case 1: Exact Phrase Search

Query: cotton shirt

The free-text VSM search ranks documents based on term weights and cosine similarity, so a document can receive a score as long as it contains the query terms, even if they are not next to each other. The positional index, however, checks the exact positions of the terms. Therefore, the phrase search only returns documents where **"cotton" is immediately followed by "shirt"**, producing a more restricted result set.

Case 2: Proximity Search

Query: cotton WITHIN/3 shirt

The proximity search uses the positions of both terms and requires **"cotton" to occur before "shirt" within 3 token positions**. Therefore, documents where the terms are farther apart, or occur in the opposite order, are excluded. This can change both the **result set and the ordering/number of matching documents** compared with a normal free-text search.

As seen from the tables above:

The cotton shirt query returns different results (DOC ID's)
for different query types:

VSM:

D001,D041,D061,D081,D022,D082,D042,D002,D062,D021

Phrase: D002,D022,D042,D062,D082

Proximity:

D001,D002,D021,D022,D041,D042,D061,D062,D081,D082